

KARNATAKA 1ST PUC BOARD PREPARATION

Chapter 5: Morphology of Flowering Plants Official Blueprint Updated Study Guide

BAIO LEARNING SYSTEM | Narendra Naslai - Founder & CEO

Subject: Biology (36) | Weightage: 5 Marks

1. OFFICIAL KARNATAKA 1ST PUC CHAPTERWISE ALLOTMENT

PARAMETER	OFFICIAL ALLOTMENT DATA (CHAPTER 5)
Chapter Name	Chapter 5 - Morphology of Flowering Plants
Teaching Hours Allotted	6 Hours (Periods)
Official Marks Allotted	5 Marks
Page Numbers in Official Notes	Pages 29 – 35
Overall Paper Format	3 Hours Max Marks: 70 Total Allotted Weightage: 105 Marks (including choices)

Expected 5-Mark Question Distribution Configurations

- **Pattern A:** 1 Five-Mark Question (Part D). **Most Common**
- **Pattern B:** 1 Three-Mark Question (Part C) + 1 Two-Mark Question (Part B).
- **Pattern C:** 1 Three-Mark Question (Part C) + 2 One-Mark Questions (Part A MCQs/VSAs).

2. TOP HIGH-YIELD TOPICS TARGETED FOR 5 MARKS ALLOTMENT

TOPIC 1 [5-Mark Essential]: Technical Description of Family Solanaceae

Frequently asked directly as a 5-Mark question in Part D: "Describe the floral characters of Family Solanaceae and write its floral formula and floral diagram."

- **Inflorescence & Flower:** Solitary, axillary or cymose (*Solanum*); Bisexual, actinomorphic.
- **Calyx:** Sepals 5, united (gamosepalous), persistent, valvate aestivation.
- **Corolla:** Petals 5, united (gamopetalous), valvate aestivation.
- **Androecium:** Stamens 5, epipetalous.
- **Gynoecium:** Bicarpellary, obligately placed, syncarpous; ovary superior, bilocular, placenta swollen with many ovules, axile placentation.
- **Floral Formula:** $\oplus \text{ } \overline{\text{K}}_{(5)} \text{ } \text{C}_{(5)} \text{ } \text{A}_5 \text{ } \text{G}_{(2)}$
- **Economic Importance:** Food (Tomato, Brinjal, Potato), Spice (Chilli), Medicine (Belladonna, Ashwagandha), Fumigatory (Tobacco), Ornamental (Petunia).

TOPIC 2: Types of Placentation with Examples (3 or 5 Marks)

Core Question: "Define placentation. Explain the different types of placentation found in flowering plants with examples."

TYPE	STRUCTURAL FEATURE	EXAMPLES
Marginal	Placenta forms a ridge along ventral suture; ovules in two rows	Pea
Axile	Placenta axial, ovules attached in a multilocular ovary	China rose, Tomato, Lemon

TYPE	STRUCTURAL FEATURE	EXAMPLES
Parietal	Ovules develop on inner wall or peripheral part; false septum present	Mustard, <i>Argemone</i>
Free Central	Ovules borne on central axis; septa absent	<i>Dianthus</i> , Primrose
Basal	Placenta develops at base of ovary; single ovule attached	Sunflower, Marigold

TOPIC 3: Aestivation Types in Corolla (3 or 5 Marks)

- **Valvate:** Sepals or petals in a whorl just touch at margins without overlapping (e.g., *Calotropis*).
- **Twisted:** One margin overlaps that of the next one in a regular direction (e.g., China rose, Lady's finger, Cotton).
- **Imbricate:** Margins overlap each other but not in any particular direction (e.g., *Cassia*, Gulmohur).
- **Vexillary (Papilionaceous):** 5 petals: largest (standard) overlaps lateral (wings), which overlap 2 smallest anterior (keel) (e.g., Pea, Bean).

TOPIC 4: Flower Position on Thalamus & Modifications (2, 3, or 5 Marks)

- **Hypogynous:** Gynoecium occupies highest position; superior ovary (e.g., Mustard, China rose, Brinjal).
- **Perigynous:** Gynoecium in centre, other parts on rim of thalamus; half-inferior ovary (e.g., Plum, Rose, Peach).
- **Epigynous:** Thalamus encloses ovary completely; inferior ovary (e.g., Guava, Cucumber, ray florets of Sunflower).
- **Monocot Seed Structure:** Presence of *Scutellum* (shield-shaped cotyledon), *Coleoptile* (protective sheath of plumule), *Coleorhiza* (protective sheath of radicle), and *Aleurone layer* (proteinaceous layer).

3. MODEL QUESTIONS MATCHED TO 5-MARK BLUEPRINT COMBINATION

Q1. What is the protective leaf-sheath covering the radicle in a monocotyledonous seed called?

1 Mark (Part A MCQ)

Model Answer: Coleorhiza. [1 Mark]

Q2. Differentiate between Racemose and Cymose Inflorescence.

2 Marks (Part B)

Model Answer:

1. **Racemose:** Main axis continues to grow endlessly; flowers borne laterally in acropetal succession. [1 Mark]
2. **Cymose:** Main axis terminates in a flower (limited growth); flowers borne in basipetal order. [1 Mark]

Q3. Differentiate between Hypogynous, Perigynous, and Epigynous flowers with one example each.

3 Marks (Part C)

Model Answer:

1. **Hypogynous:** Gynoecium at highest position, ovary superior. E.g., Mustard / China rose. [1 Mark]
2. **Perigynous:** Gynoecium in centre, ovary half-inferior. E.g., Rose / Peach. [1 Mark]
3. **Epigynous:** Thalamus encloses ovary completely, ovary inferior. E.g., Guava / Cucumber. [1 Mark]

Q4. Describe the family Solanaceae with respect to floral characters, floral formula, and economic importance.

5 Marks (Part D)

Model Answer:

1. **Calyx & Corolla:** Sepals 5, united, persistent, valvate; Petals 5, united, valvate. [1 Mark]
2. **Androecium:** Stamens 5, epipetalous (attached to petals). [1 Mark]
3. **Gynoecium:** Bicarpellary, syncarpous, superior ovary, bilocular with swollen placenta and axile placentation. [1 Mark]
4. **Floral Formula:** $\oplus \text{ } \overline{\text{K}}_{(5)} \text{ } \overline{\text{C}}_{(5)} \text{ } \overline{\text{A}}_5 \text{ } \overline{\text{G}}_{(2)}$. [1 Mark]
5. **Economic Importance:** Food (Potato/Tomato), Medicine (Ashwagandha/Belladonna), Spice (Chilli). [1 Mark]

BLUEPRINT SCORING TIP FOR KARNATAKA 1ST PUC

Since Chapter 5 carries **5 Marks** with **6 Teaching Hours**, master the complete 5-mark technical description of Family Solanaceae or the 5 types of placentation along with clear definitions of floral whorl arrangements.